

B. TECH.
(SEM VI) THEORY EXAMINATION 2022-23
EMBEDDED SYSTEM

Time: 3 Hours**Total Marks: 100****Note: 1.** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief. 2 x 10 = 20

- a. List out the challenges in building an embedded system.
- b. How SPI is different from other serial interfaces?
- c. Discuss the functions of RTOS?
- d. List the different phases of EDLC.
- e. Explain state machine model?
- f. Mention the different states of thread.
- g. Discuss remote procedure call?
- h. Define Sensors and Actuators.
- i. Discuss watchdog timer?
- j. Discuss shared memory.

SECTION B

2. Attempt any three of the following: 10 x 3 = 30

- a. List and brief the main characteristics of embedded systems that distinguish such systems from other computing systems.
- b. Explain the terminologies semaphores, pipes and shared memory in RTOS.
- c. Provide various communication strategies for embedded systems.
- d. With suitable diagram explain in detail about the concept of washing machine application.
- e. Explain the memory management. Also explain working of DMA.

SECTION C

3. Attempt any one part of the following: 10 x 1 = 10

- a. Discuss Oscillator unit. Explain the role of real time clock in embedded system.
- b. Discuss serial bus communication protocols. Also discuss its use in networking of embedded systems.

4. Attempt any *one* part of the following: 10 x 1 = 10

- a. Explain the product life-cycle curve of an embedded product development.
- b. Discuss the advantages and disadvantages of edge-triggered and level-triggered interrupts?

5. Attempt any *one* part of the following: 10 x 1 = 10

- a. Discuss the various techniques of task synchronization.
- b. Describe timing and clock in embedded system with relevant example.

6. Attempt any *one* part of the following: 10 x 1 = 10

- a. Explain various families of ARM Processors. Also mention typical features of each.
- b. Discuss the role of RAM and ROM in embedded system. Explain.

7. Attempt any *one* part of the following: 10 x 1 = 10

- a. Brief the issues of real time operating systems.
- b. Give the brief content of the following terms:
 - i) Serial Peripheral Interface
 - ii) Inter Integrated Circuits